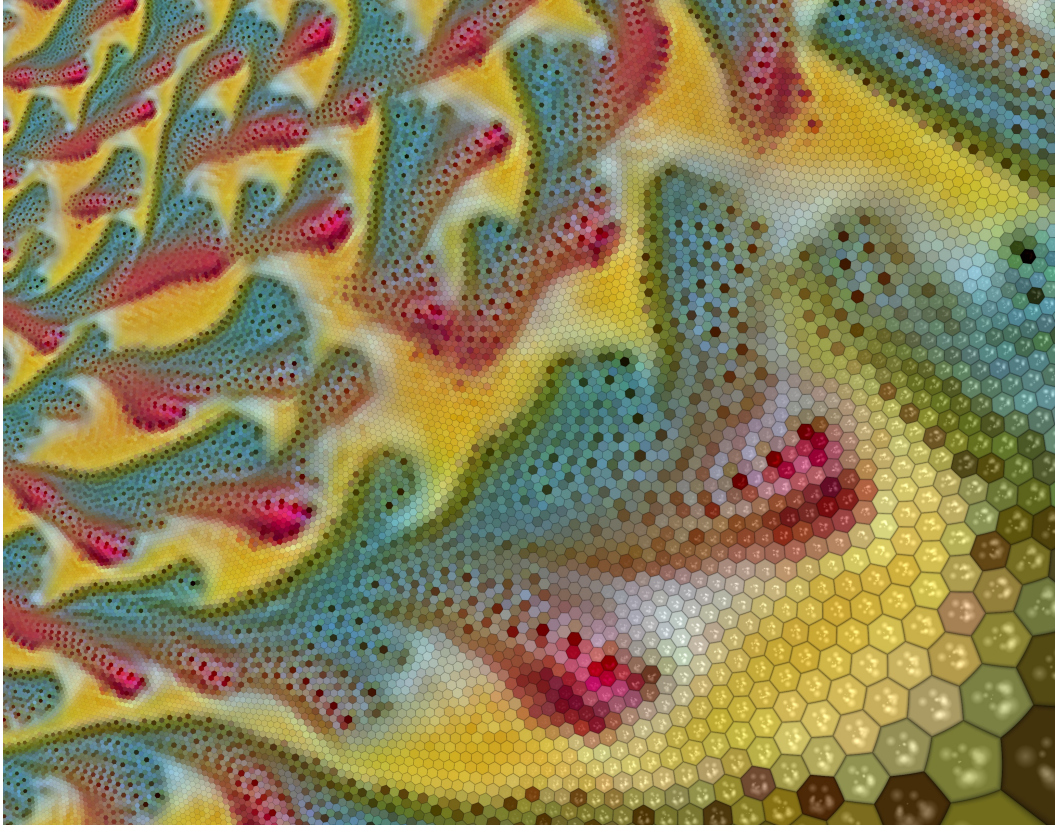


Hexells: Self-Organizing Textures

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PROJECT URL

<https://znah.net/hexels>

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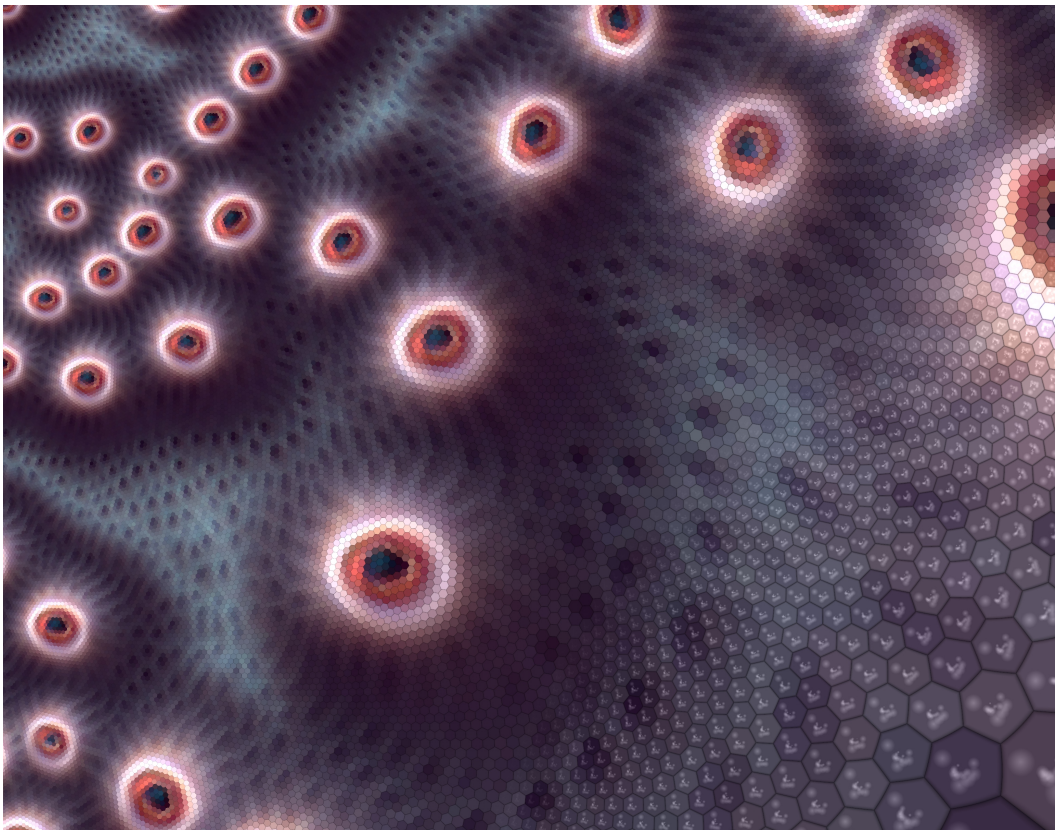
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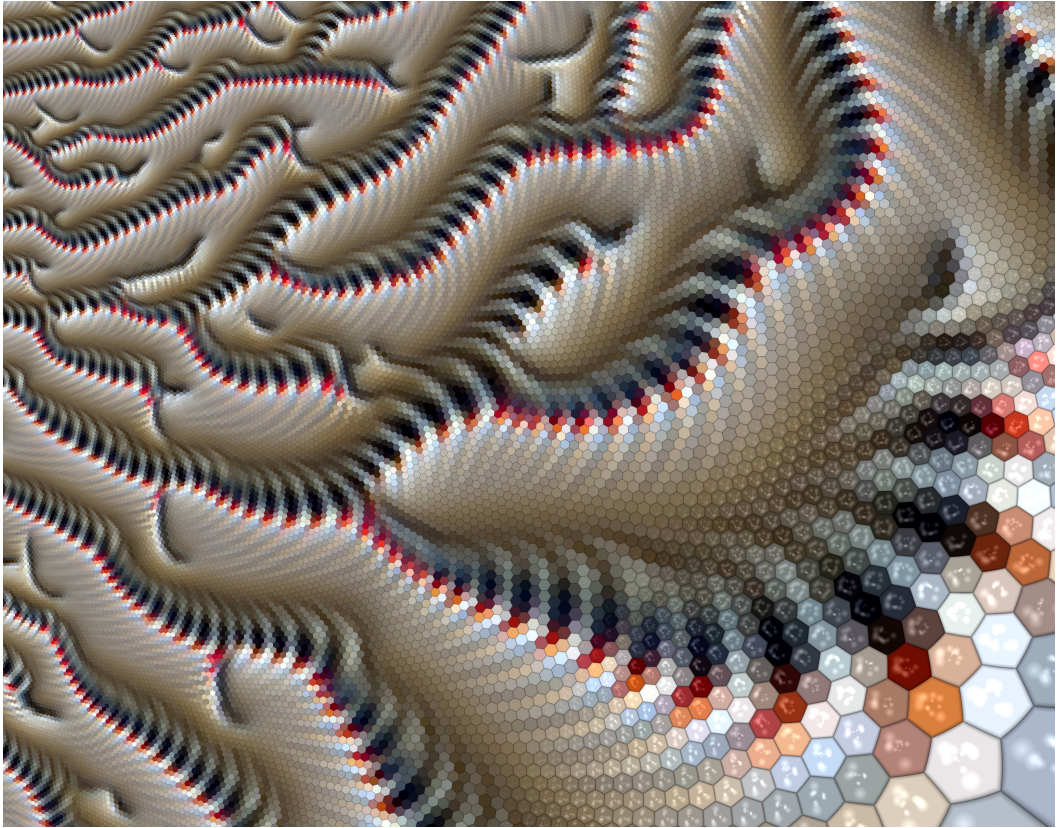
DESCRIPTION OF ARTWORK

Hexells is a self-organizing system of cells that was designed to synthesize and maintain various textures through local communication between the cells. It is inspired by natural pattern formations found on plants and animal skins. The cellular automata (CA) update rule is modeled using a small neural network inspired by Mordvintsev et al. [2020]. We created a WebGL demo that allows users to explore and interact with more than one hundred learned CA texture-generating behaviors. The chosen projection shows the system behavior at local and global scales at the same time.

We exploit two types of training objectives for training the CA rules. One is based on the example-based texture synthesis objective [Gatys et al. 2015], which tries to learn a behavior that would produce a “camouflage” pattern that matches a provided texture sample. Another objective is based on feature visualization [Olah et al. 2017], which tries to create patterns that excite individual neurons in the visual system of another animal (modeled using the ImageNet-pretrained convolutional network). This objective has parallels with the deceiving camouflage used by some creatures to scare away predators or attract prey.

Hexells harnesses the power of self-organization that is omnipresent in nature. The created system is capable of achieving a global shared goal of pattern synthesis through the local communication between individual agents executing the same rule.





ARTIST INFORMATION

Alexander Mordvintsev is a researcher and an artist working with machine perception, self-organization, and artificial life. He is known for developing the *DeepDream* machine hallucination algorithm. <https://znah.net>

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